

Problem Set 7

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Problem 1.

Using the short-run goods-market setup:

$$D = C(Y - T) + I + G + \text{CA}\left(\frac{EP^*}{P}, Y - T\right), \quad Y = D\left(\frac{EP^*}{P}, Y - T, I, G\right).$$

In the short run, P and P^* are fixed.

(a) Goods market (SR)

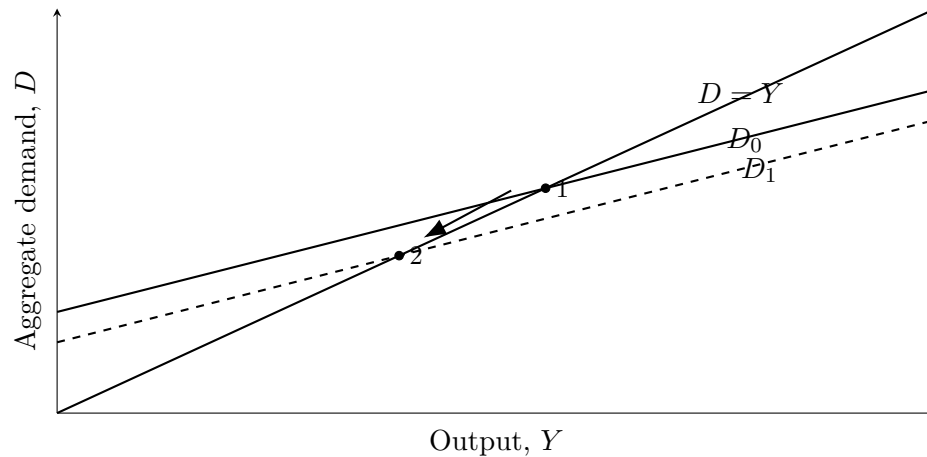
A tax increase reduces disposable income:

$$T \uparrow \Rightarrow (Y - T) \downarrow \Rightarrow C(Y - T) \downarrow.$$

Lower disposable income also reduces imports, which tends to improve the current account:

$$(Y - T) \downarrow \Rightarrow \text{CA}\left(\frac{EP^*}{P}, Y - T\right) \uparrow.$$

Overall, aggregate demand falls at a given Y , so the D schedule shifts down/left and equilibrium output falls.

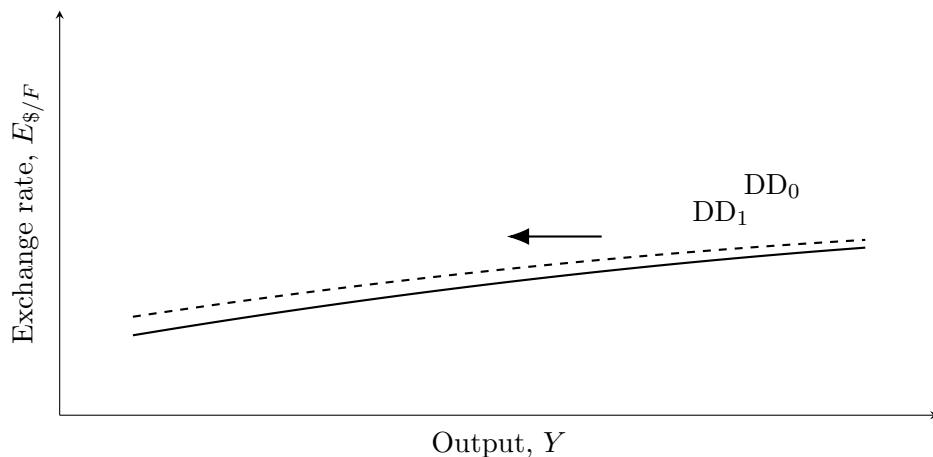


(b) DD schedule (SR)

DD is the set of (Y, E) pairs such that $Y = D()$ holds in the short run. With P and P^* fixed, an increase in $E_{\$/F}$ raises the real exchange rate $\frac{EP^*}{P}$, improves the current account, and increases demand and output.

A tax increase reduces demand at any given (Y, E) , so DD shifts left:

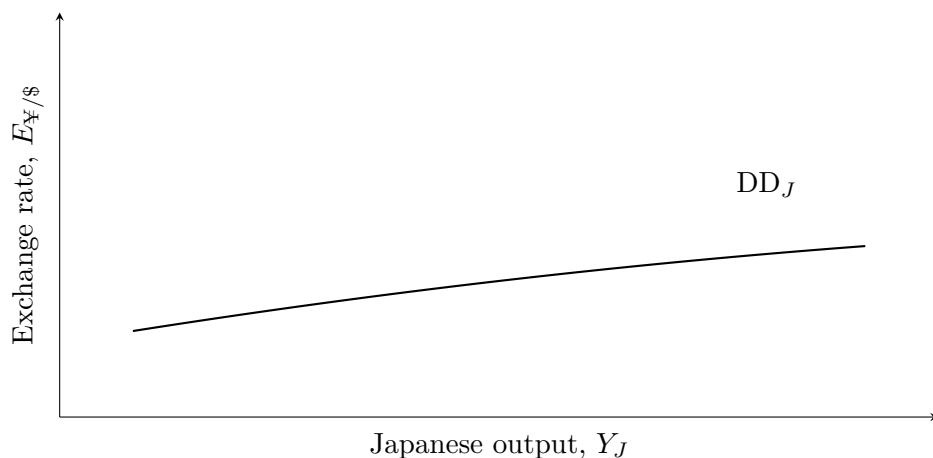
$$T \uparrow \Rightarrow D \downarrow \Rightarrow \text{DD}_0 \text{ shifts left to } \text{DD}_1.$$



Problem 2.

(a) DD schedule for Japan.

With short-run prices fixed, a rise in $E_{¥/\$}$ is a yen depreciation, which raises $\frac{EP^*}{P}$ and improves the current account, increasing demand and output.



(b) For each change, shift DD LEFT, RIGHT, or NO SHIFT.

$$D = C(Y - T) + I + G + \text{CA}\left(\frac{EP^*}{P}, Y - T\right).$$

Any exogenous change that increases (decreases) demand for home goods shifts DD right (left).

(i) **Increase in Japanese government spending G :**

$$G \uparrow \Rightarrow D \uparrow \Rightarrow \text{DD shifts RIGHT.}$$

(ii) **Japanese consumers want to consume less/save more:**

$$C() \downarrow \Rightarrow D \downarrow \Rightarrow \text{DD shifts LEFT.}$$

(iii) **Exogenous increase in prices in Japan:**

$$P \uparrow \Rightarrow \frac{EP^*}{P} \downarrow \Rightarrow \text{CA} \downarrow \Rightarrow D \downarrow \Rightarrow \text{DD shifts LEFT.}$$

(iv) **Exogenous increase in the exchange rate $E_{¥/\$}$:** Since $E_{¥/\$}$ is the axis variable, this is a **movement along DD (NO SHIFT)**.

(v) **Foreign sanctions reduce foreigners' propensity to buy Japanese goods:** Exports fall at any real exchange rate, so CA shifts down:

$$\text{CA} \downarrow \Rightarrow D \downarrow \Rightarrow \text{DD shifts LEFT.}$$